Project 1 Phase II: Design for Deflection Report

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Executive Summary

In Phase II of our project, we designed a structure intended to deform linear elastically under a maximum vertical load of 100 lbf and produce a target deflection of 0.4 inches at the load point, corresponding to a spring constant of 250 lbf/in.We used Solidworks' Finite Element Analysis (FEA) feature in a static simulation to iteratively design a "C-shaped" structure to fit on a 4" x 8" maximum plate area with mounting holes in fixed positions to allow for instron testing and 0.25" thickness. We prioritized achieving the desired stiffness and not exceeding the elastic limit of our material (Aluminum 6061-T6) while minimizing weight to increase efficiency and decrease cost. We refined our geometry in CAD based on each FEA prediction until we achieved results we were confident in. After testing with the Instron, the measured deflection (0.4408 in) closely matched our FEA prediction (0.4 in) and the structure deformed linear elastically without yielding. Overall, our part was slightly compliant with a spring constant of 228.5 lbf/in. Since our aluminum thickness ended up being slightly below 0.25" (0.239"), correcting for this would likely improve our spring constant significantly.

Introduction

In completing Phase II of this project, our goal was to design an efficient structure that would deform linear elastically with a deflection as close to .4 inches as possible with minimal yielding. An efficient structure is one that hits the target performance with the least material, thus, a lighter structure is optimal. We knew that there would be a maximum vertical force of 100 lbf applied by the Instron machine, so we designed our structure to have its maximum deflection at the force's point of application. We started designing 3 different types of structures and subsequently tested each one to see which would hit our deflection and yield goals the best. In order to be tested using the instron machine, our design was constrained to a maximum plate area of 4" by 8" and needed three mounting holes in assigned positions with protected areas that cuts could not encroach on. For our simulation, we assumed the thickness of the aluminum to be 0.25 inches.

We began the design process with two basic shapes: a C-shaped structure and an S-shaped structure. Much of designing entailed making minor changes to these structures by adding or removing material or altering the shape to best match the desired deflection and minimize yielding. After each change to the structures, re-meshed the structure and repeated the static FEA simulation to find the deflection at the point of load application and analyze sites prone to yielding using the Von-Mises stress contour plot. To accurately assess yielding in vulnerable areas, we applied a finer mesh in those areas for more accurate stress readings there. After optimizing structures for the desired deflection with minimal yielding, we compared our different shaped designs based on weight. Ultimately, this led to our decision to use the C-shaped design since it was efficient in terms of weight while still meeting deflection goals without significant yielding.

Analysis

Geometry

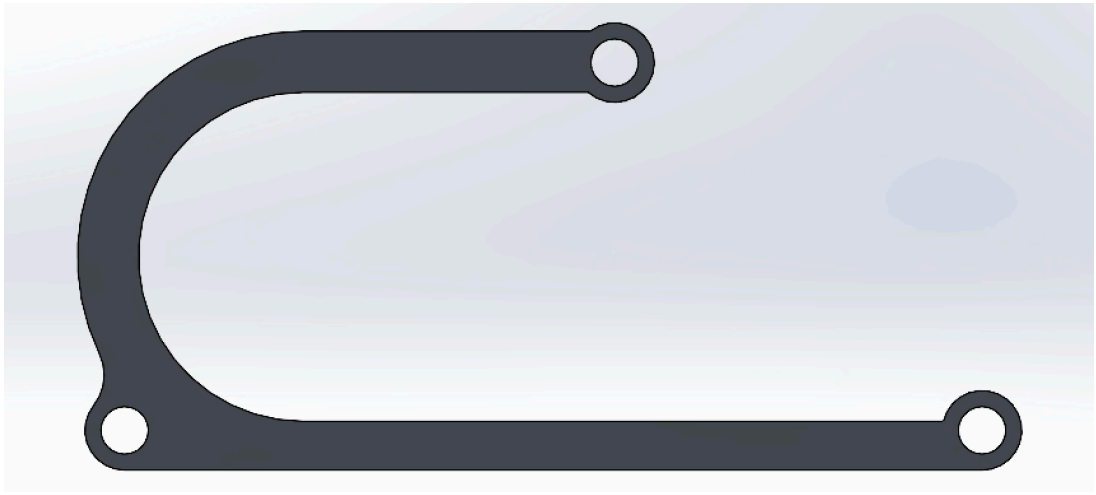


Figure 1: Front view of C-shaped design

Displacement

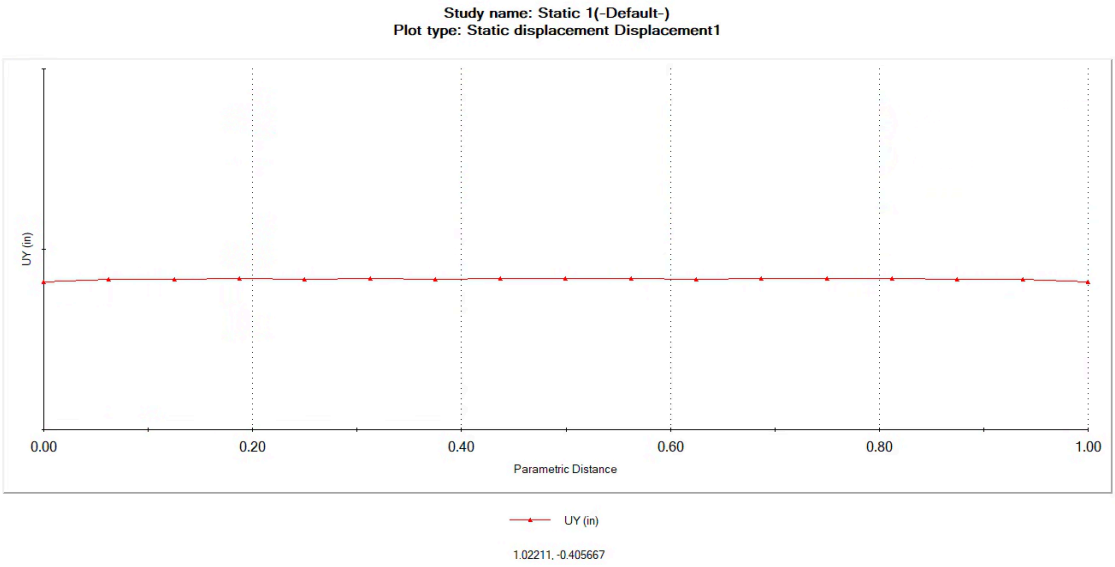


Figure 2: Vertical deflection in inches probed along the split line of load application (SolidWorks)

Figure 2 shows the vertical displacement (Uy) in inches along a defined path on the structure under the applied 100 lbf load. The plot was obtained by probing for Uy along the split line where the load is applied. The x-axis represents the parametric distance, which corresponds to the normalized position along the path (from 0 to 1), while the y-axis shows the vertical deflection in inches. This plot reveals the predicted maximum deflection at the site of load application to be **0.405667 in** (actual deflection of **0.4408 in** during Instron testing)

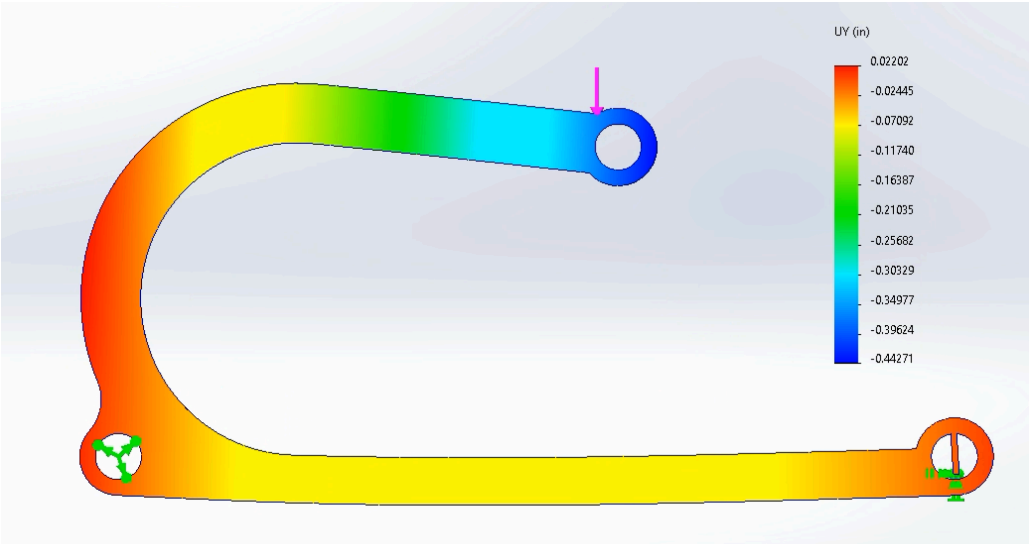


Figure 3: Colored contour plot of vertical displacement (Uy) in inches (SolidWorks)

Figure 3 shows the colored contour plot of the vertical displacement in inches, revealing locations that are most displaced. We can see that the site of application of the load (the bottom midline of the top mounting hole) is displaced the most (indicated in blue, note we are looking at downward displacement, which is negative). This displacement decreases as you move rightward from the load (mirroring a cantilevered beam). Additionally, we can see some downward displacement in the center of the bottom length of the structure due to bending, showing how our simulation allows motion of the right mounting point using a strut. This further displaces the site of load application.

Stress

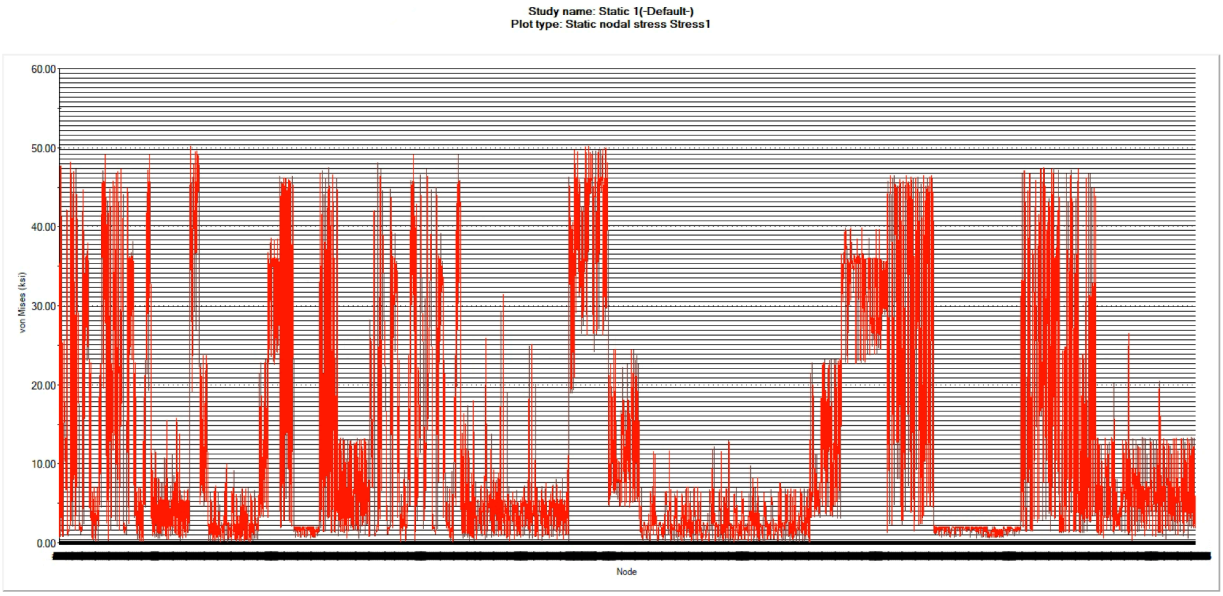


Figure 4: Von Mises stress (ksi) probed on all faces of the surface (excluding faces affected by the strut) (SolidWorks)

Figure 4 shows the Von Mises stress at each node probed on all surfaces of the structure. The faces of the structure containing the strut were excluded since stress is high here to allow for bending of the bottom length. The Instron machine applies the desired simple support at this location without using a strut, so the stress in this region is neglected. This plot reveals that the maximum Von Mises stress in the structure is **50.171 ksi**. Note that the yield strength of Aluminum 6061-T6 is approximately 39.885 ksi.

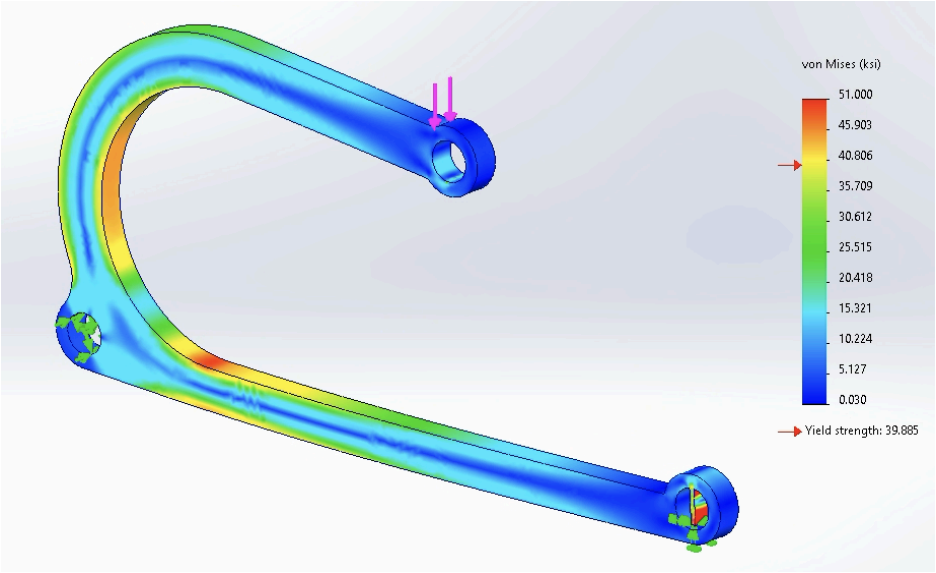


Figure 5: Colored contour plot of Von Mises stress in ksi.

Figure 5 reveals the locations of the sites of high stress in the structure. The maximum Von Mises stress resides at the connection between the curve of the C and the bottom length of the structure. Other notable high-stress regions are at this location at the bottom of this bottom length and where the curve of the C meets the protected region above the left mounting hole.

So, the maximum stress in the structure, 50.171 ksi, is around 10 ksi above the yield stress of Aluminum 6061-T6 (39.885 ksi), so we would expect some yielding under the load which may inhibit linear elastic behavior. Compared to the other geometries we tested using FEA, this was the least amount of stress we were able to achieve, indicating our structure distributes stress more effectively than other structures.

Mesh Analysis

While running and adjusting the simulations to best meet displacement requirements while minimizing stress, one parameter that affected the accuracy of the results was the quality of the mesh. To run simulations quickly, we began by using a mesh density intermediate between the course and fine meshes without any refinement. This granted us a baseline understanding of how the structure performed. If the results were undesirable, the structure was modified before further simulations were completed. If the results seemed more promising in terms of balancing high deflection (near 0.4 in) and low stress, we added mesh controls around the mounting holes where the material is thin and prone to stress and where there are sharp bends and corners in the structure, making the mesh fine in these locations. This was preferable to making the mesh fine everywhere to optimize computing time. The results were virtually unchanged after refining the mesh.

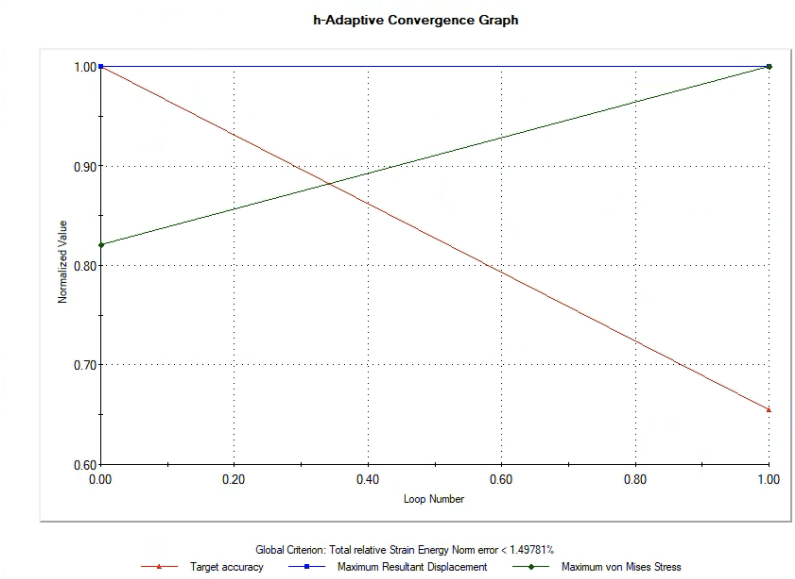
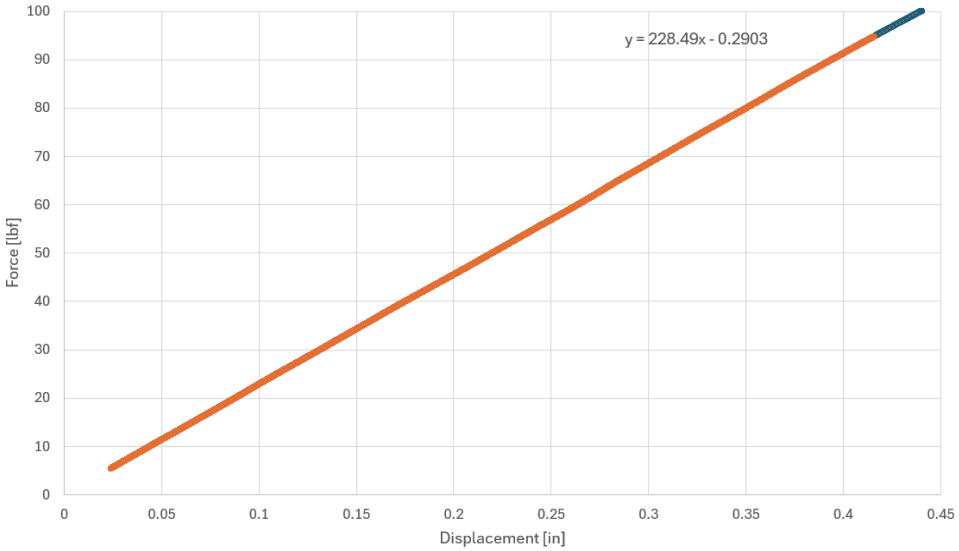


Figure 6: h-adaptive convergence graph (SolidWorks)

Additionally, we ran a mesh refinement study (h-adaptive), which progressively refines the mesh and re-runs the simulation. The h-adaptive test stopped running after only one loop since our results were within the target accuracy 98% (98.5022% accurate) without needing to re-run. Thus, it is safe to assume that our result has converged, indicating that the results are relatively independent of the applied mesh (i.e. stable results, additional mesh refinement would not change the results significantly). This can be seen in Figure 6, where the maximum resultant displacement is relatively constant (no change with refinement) and the maximum von mises stress increases only slightly, indicating stresses become slightly higher with refinement (maximum 50.228 ksi).

Testing Results



Spring Constant (lbf/in)	228.5 lbf/in
Weight	68 g
Average plate thickness	0.239 in
Yielding	None
Max. Deflection	0.4408 in

Figure 7: Load-deflection test data from the Instron machine. Force in lbf plotted as a function of displacement in inches. The line of best fit

Our structure fared well when considering the performance metrics. Through Instron testing, we measured the deflection to be 0.4408 in at the load point when the maximum 100 lbf is applied, closely matching the requirement of 0.4 in (10.2% error). The structure deformed linear elastically without yielding. Overall, our part was slightly more compliant than predicted with a spring constant of 228.5 lbf/in, about 8.6% less than the ideal spring constant of 250 lbf/in. This deviation is likely due to manufacturing tolerances, fixture compliance, and small differences in material thickness (simulation used 0.25 in thick aluminum while true average thickness was 0.239 in), all of which can slightly reduce stiffness. We ended up re-running the simulation with 0.239 in aluminum which returned a maximum displacement of 0.42511 in, which is much closer to the 0.4408 in measured result (3.5% error). This indicates that aluminum that was 0.25 in thick would have improved the spring constant and better matched the deflection returned by the simulation initially (0.405667 in). Overall, our structure was extremely efficient, weighing only 68 g. Adjusting for material thickness would likely make the spring constant value very close to 250 lbf/in. Both simulation and Instron testing validated the effectiveness of our design process and confirmed that the structure met the performance goals with minimal excess material.

Discussion

The primary trade off that we made when optimizing for material efficiency was settling with a slightly higher stress than was ideal in our FEA iterations. The yield strength of our aluminum is approximately 39.885 ksi. Our primary difficulty was removing as much material as possible while not dipping below the optimal spring constant of 250 lbf/in. We had two final designs: one with a stress of 47 ksi and deflection slightly under 0.4 in and another with a stress of 49 ksi and a deflection of exactly 0.4 in. We ended up choosing the latter because it weighed less, even though it had a slightly larger stress value. We reasoned that the difference in stress was a small enough sacrifice and that optimizing for weight and hitting the exact deflection was the more important priority. Though our FEA predicted stress value was greater than the yield strength of our aluminum, the material did not yield. This deviation between the predicted (FEM) and observed results (Instron) was likely due to differing real-world material properties. One possibility is that the grain structure was aligned just right, enhancing local strength beyond the nominal yield value. Furthermore, factors such as surface finish, residual stresses from machining, and the conservatism of the FEA mesh could have contributed to the discrepancy. The real material may also have experienced slight strain hardening under load, effectively increasing its yield strength. Overall, these effects likely combined to produce a stiffer outcome than the FEA predicted.

The other deviation in our experiment was that our observed deflection result of 0.4408 in was slightly larger than our predicted result of 0.405667 in and the desired 0.4 in. This discrepancy can be attributed to a few factors. Our FEA model assumed idealized boundary conditions and perfect rigidity at the constraints, while in reality, there was likely some compliance in the fixture and loading setup that allowed for greater displacement. Other causes are likely small tolerances in the machining and assembly, such as imperfect alignment, minor gaps at the supports, and inconsistent material thickness. Furthermore, the FEA mesh size may have slightly influenced stiffness predictions, especially near high stress concentration. Also, slight material differences or inconsistencies in modulus of elasticity or internal microstructure could have caused a more compliant response. Overall, the measured deflection remained within a close margin of error ((0.4-0.4408)/(0.4) * 100 = ~ 10.2%), suggesting that our FEA model accurately captured the system's overall mechanical behavior despite these small real-world effects.